



CSE 4621: Microprocessors and Interfacing

Lecture 2 — Overview of Microcomputer Structure and Operation

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Lecture Plan

The agenda for today

- Structure of a Microcomputer (*in more detail*)
- Design choices of different components (*in more detail*)
- Revisit the Fetch–Decode–Execute Cycle (*in more detail*)

Microcomputer

What is it?

A microcomputer is a small, relatively inexpensive computer having a *central processing unit (CPU)* made out of a single integrated circuit microprocessor.

- also includes *memory* and *input/output (I/O) circuitry* together mounted on a printed circuit board (PCB).

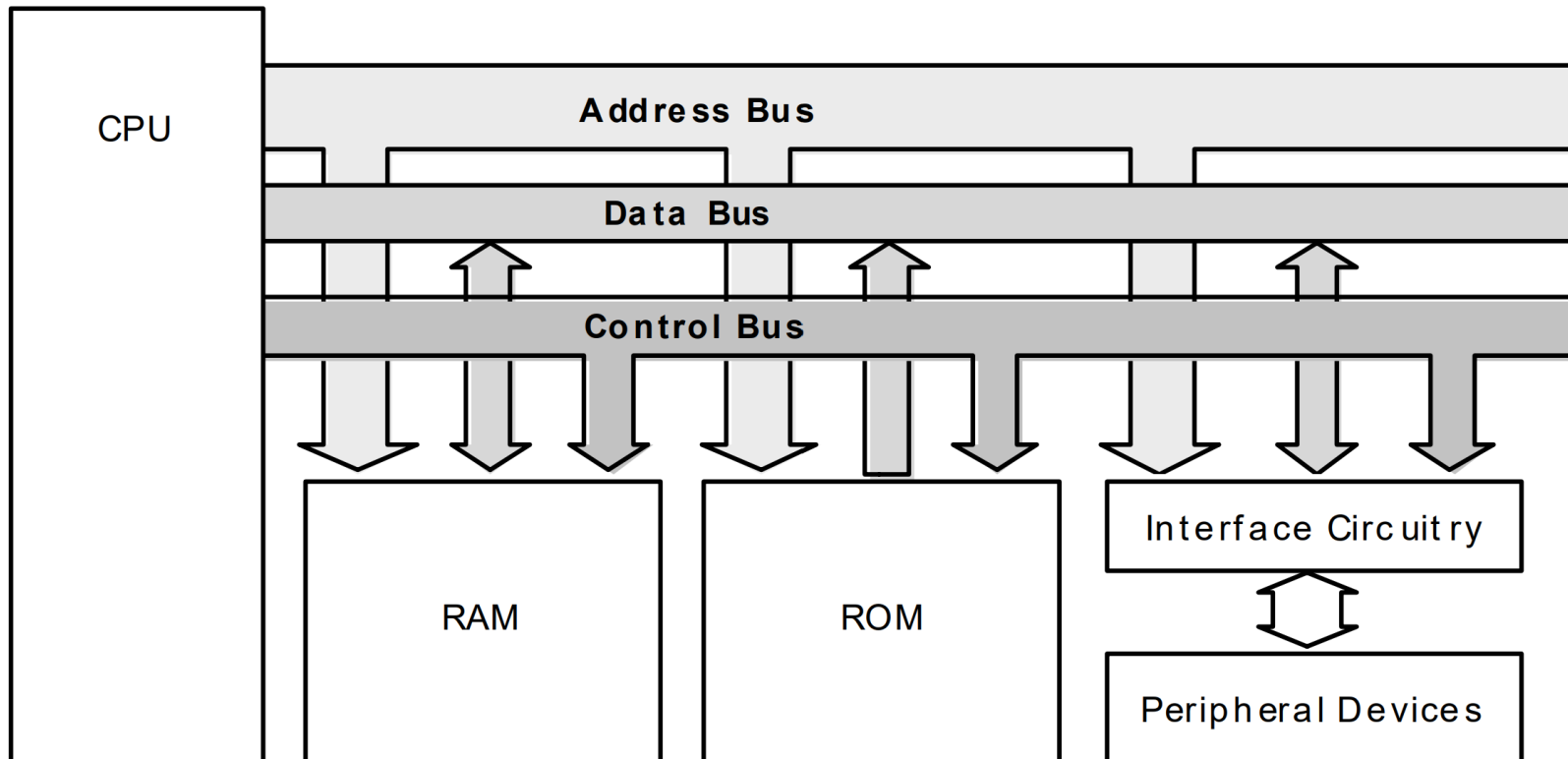


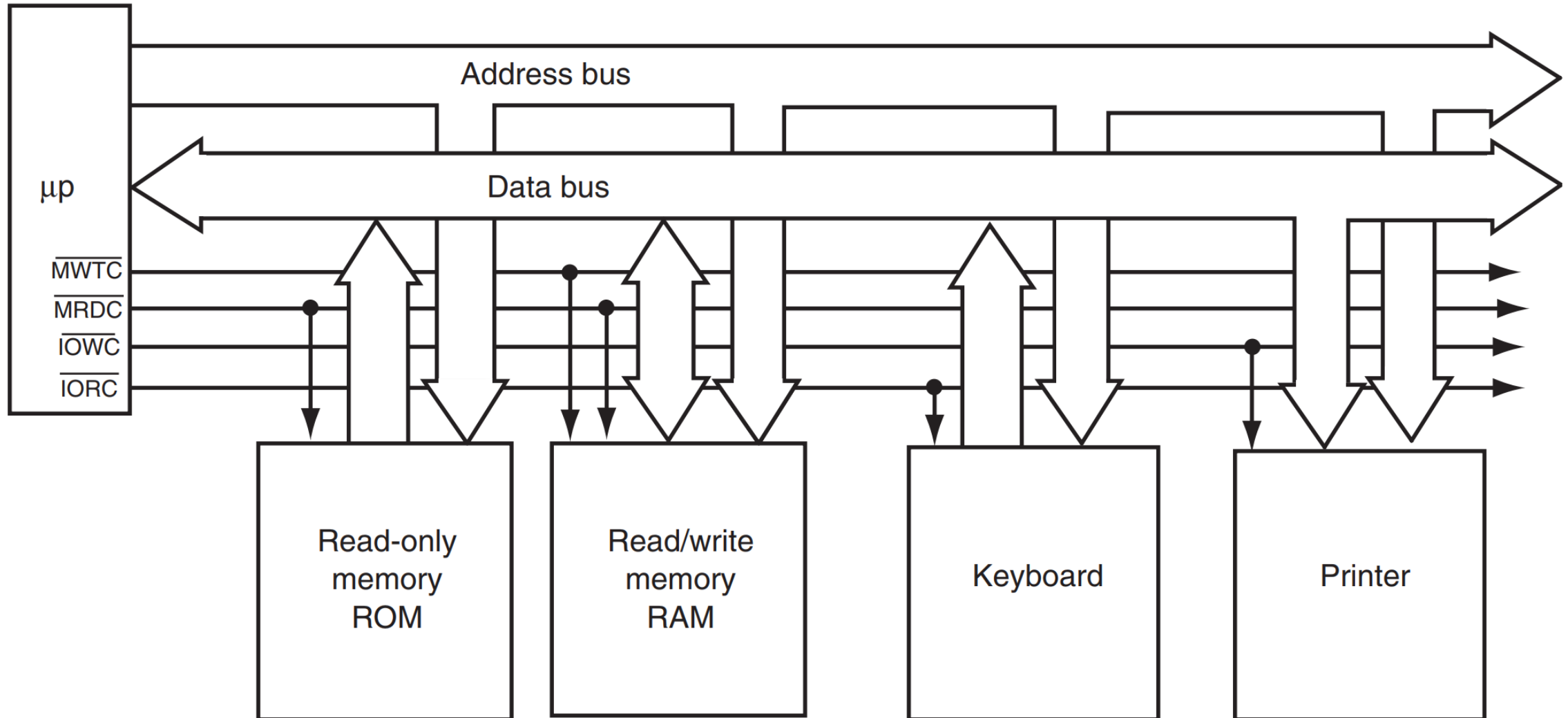
Fig: Microcomputer Block Diagram



Fig: Raspberry Pi, a popular modern-class microcomputer

Microcomputer

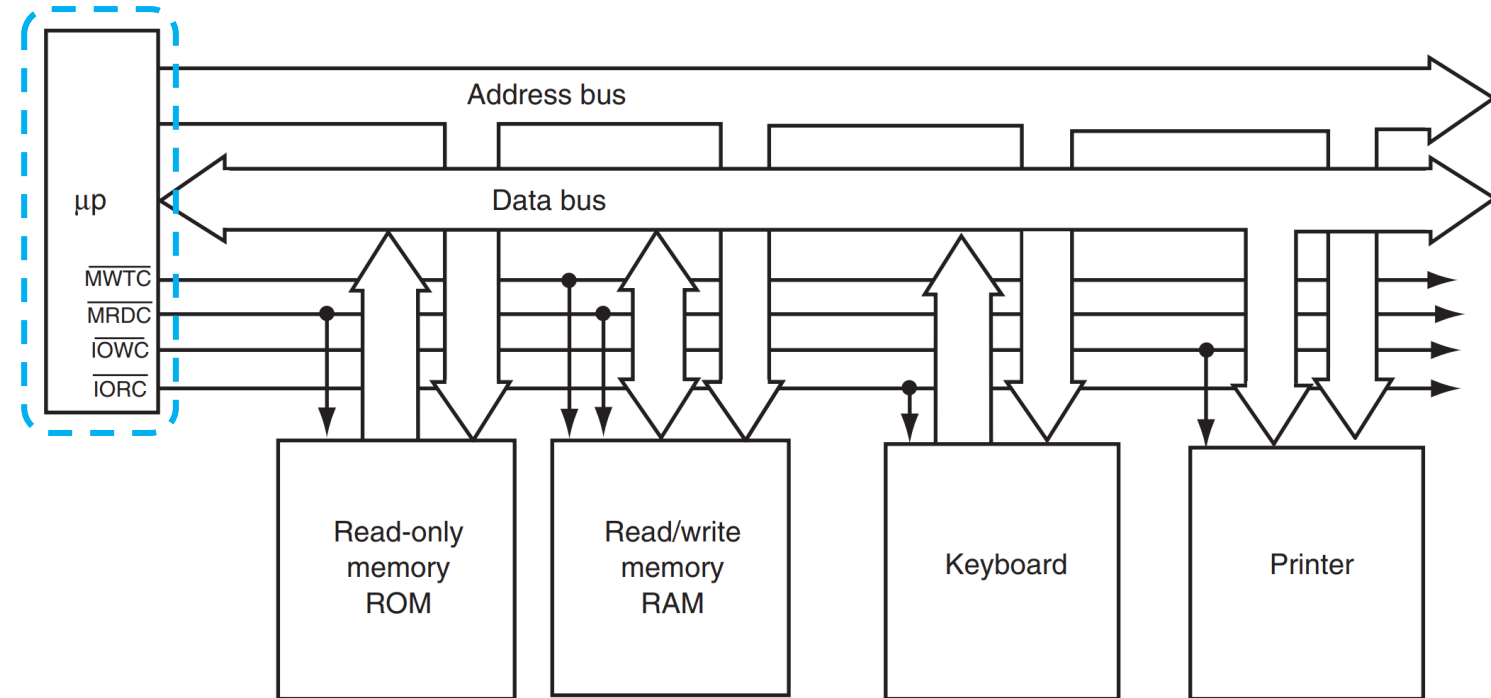
Block diagram magnified



Microcomputer

Central Processing Unit (CPU)

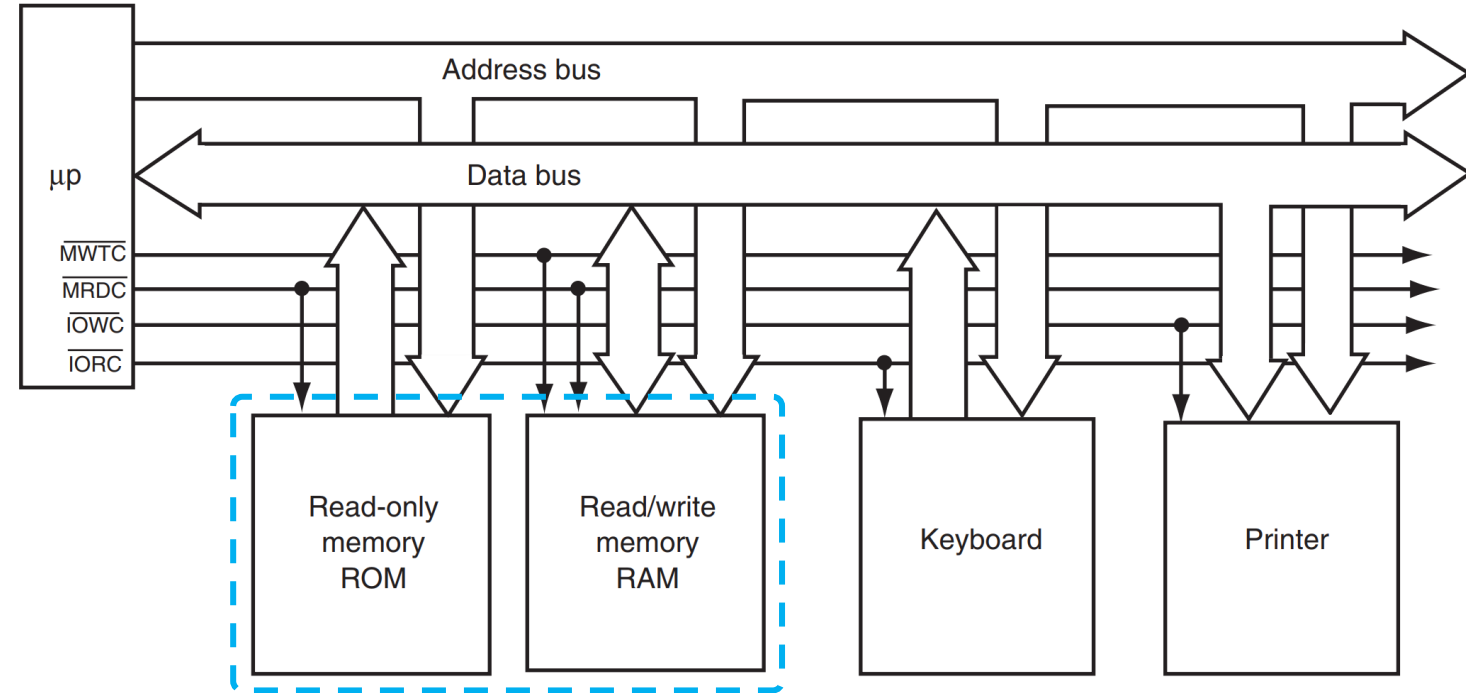
- It *controls* the operation of computer
- The CPU *fetches* binary-coded instructions from memory
- *Decodes* the instructions into a series of simple actions
- *Carries out* these actions in a sequence of steps
- **Important Components:**
 - ✓ IP (Instruction Pointer),
 - ✓ General purpose registers, and
 - ✓ Control bus signal generating circuits



Microcomputer

Memory

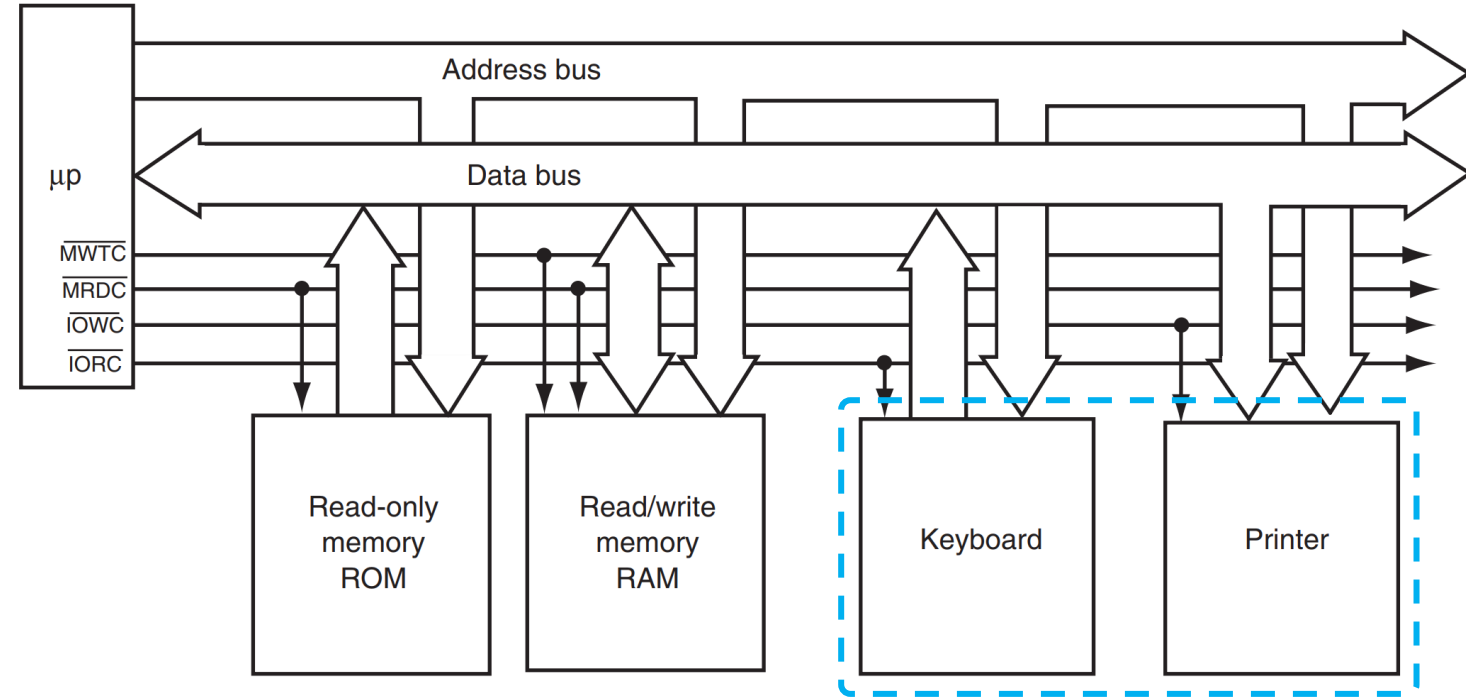
- **Stores** the binary codes for the sequence of **instructions** and binary coded **data**.
- **RAM** can be **read** and **written** to anytime.
 - ✓ Erased when the power to the computer is turned off (**volatile**)
- **ROM** is pre-loaded with data and software that **never changes**, so the CPU can **only read** from it.
 - ✓ Used to store the computer's initial start-up instructions
 - ✓ Retains its data indefinitely (**non-volatile**)
- **BIOS** **orchestrates** loading the computer's operating system from the hard disk drive into RAM whenever the computer is turned on or reset.



Microcomputer

Input/Output (I/O)

- Refers to the communication between an information processing system (*e.g.*, **computer**), and the outside world possibly a **human**, or another information processing system.
- **Inputs** are the signals or data received by the system.
- **Outputs** are the signals or data sent from it.
- Devices that provide input or output to the computer are called **peripherals**, *e.g.*, keyboard, mouse, printer, display, etc.



Microcomputer

System Bus

- Bus → bundle of wires
- The microcomputer's system bus contains three buses, **address**, **data**, and **control** bus.
- When a memory or an I/O chip receives data from the microprocessor, it is called a **WRITE** operation, and data is written into a selected memory location or an I/O port (register).
- When a memory or an I/O chip sends data to the microprocessor, it is called a **READ** operation, and data is read from a selected memory location or an I/O port.

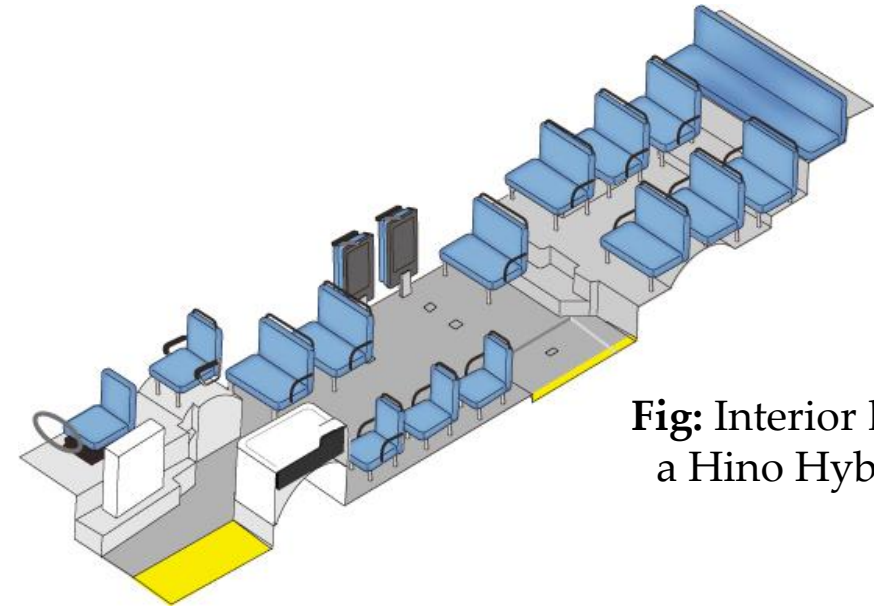
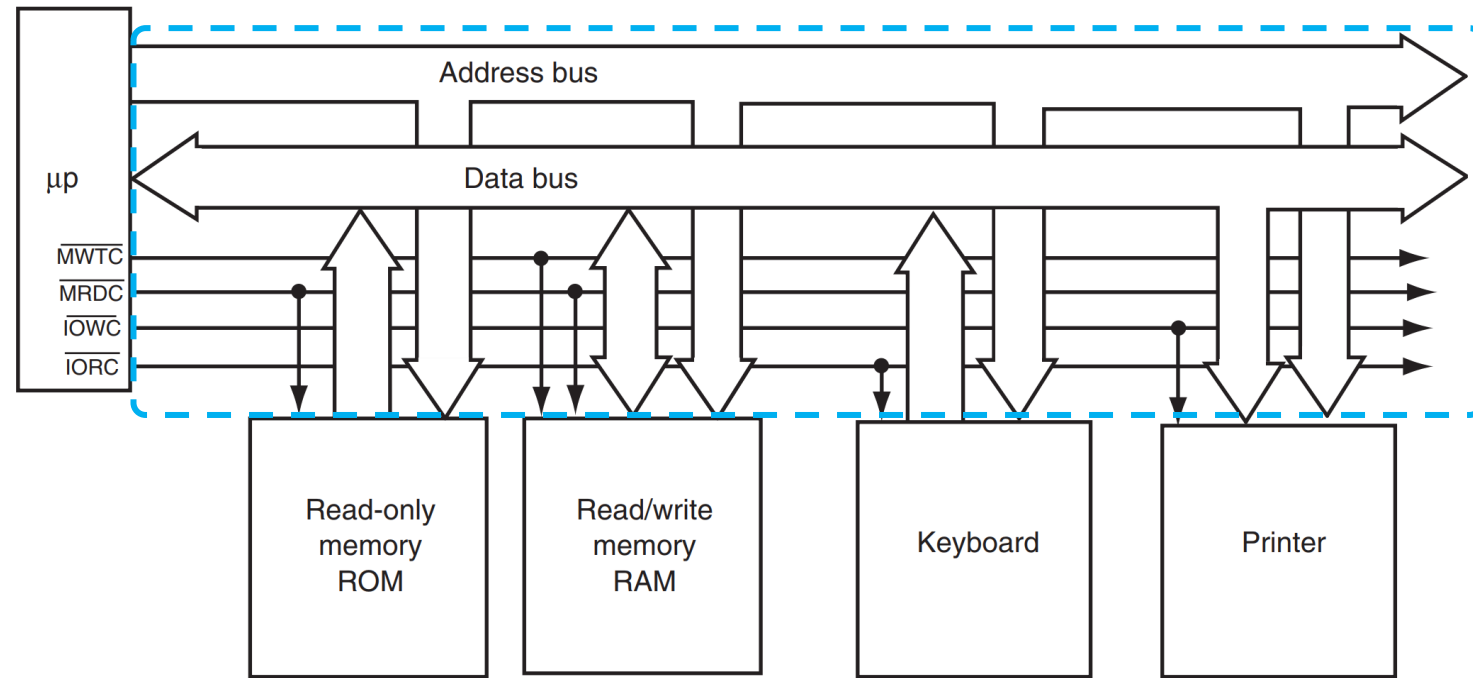
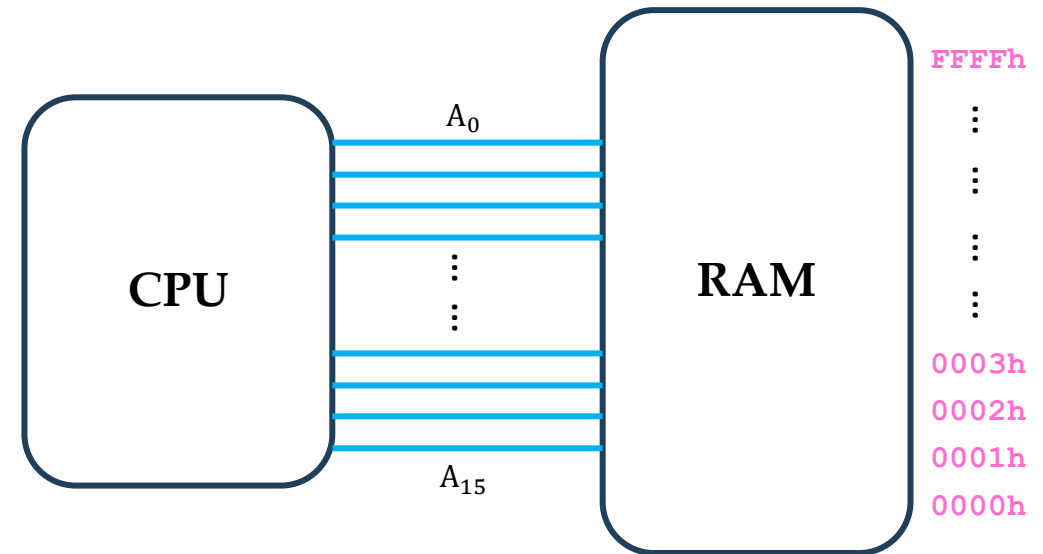
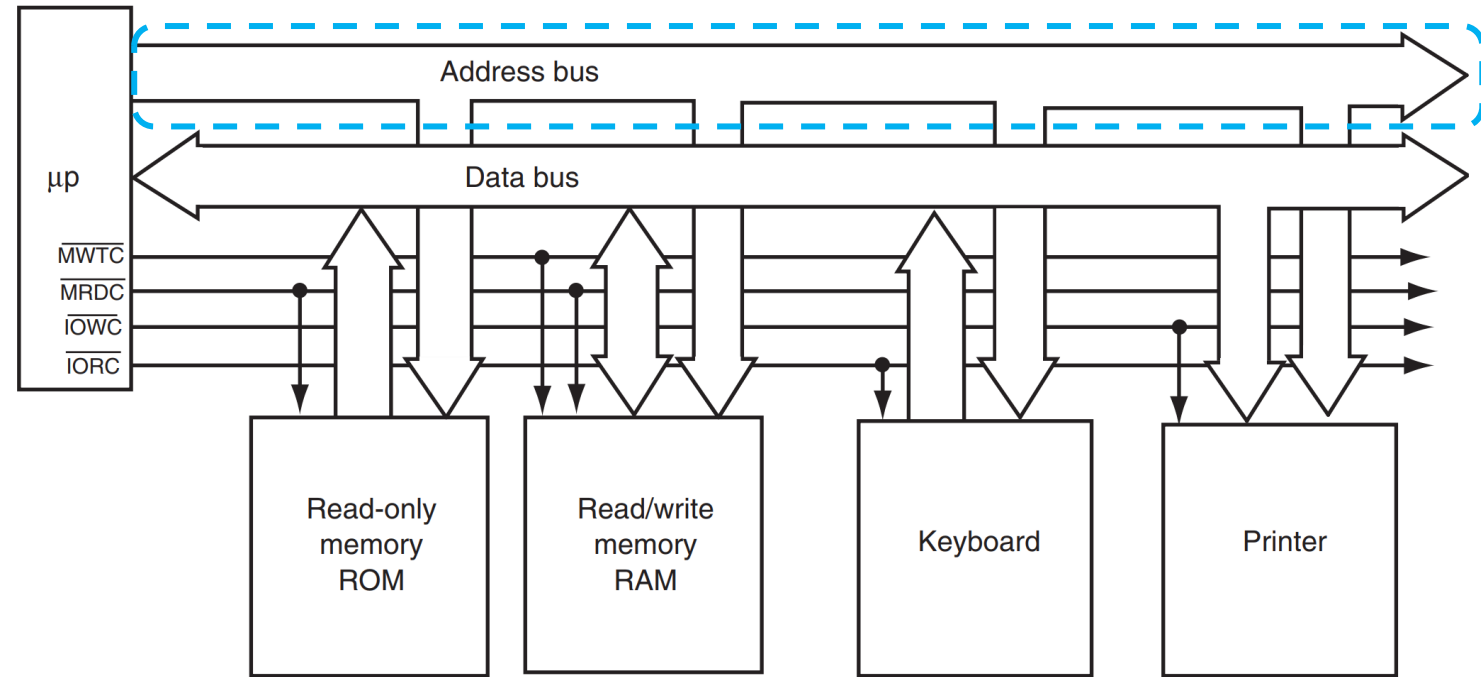


Fig: Interior layout of a Hino Hybrid bus

Microcomputer

Address Bus

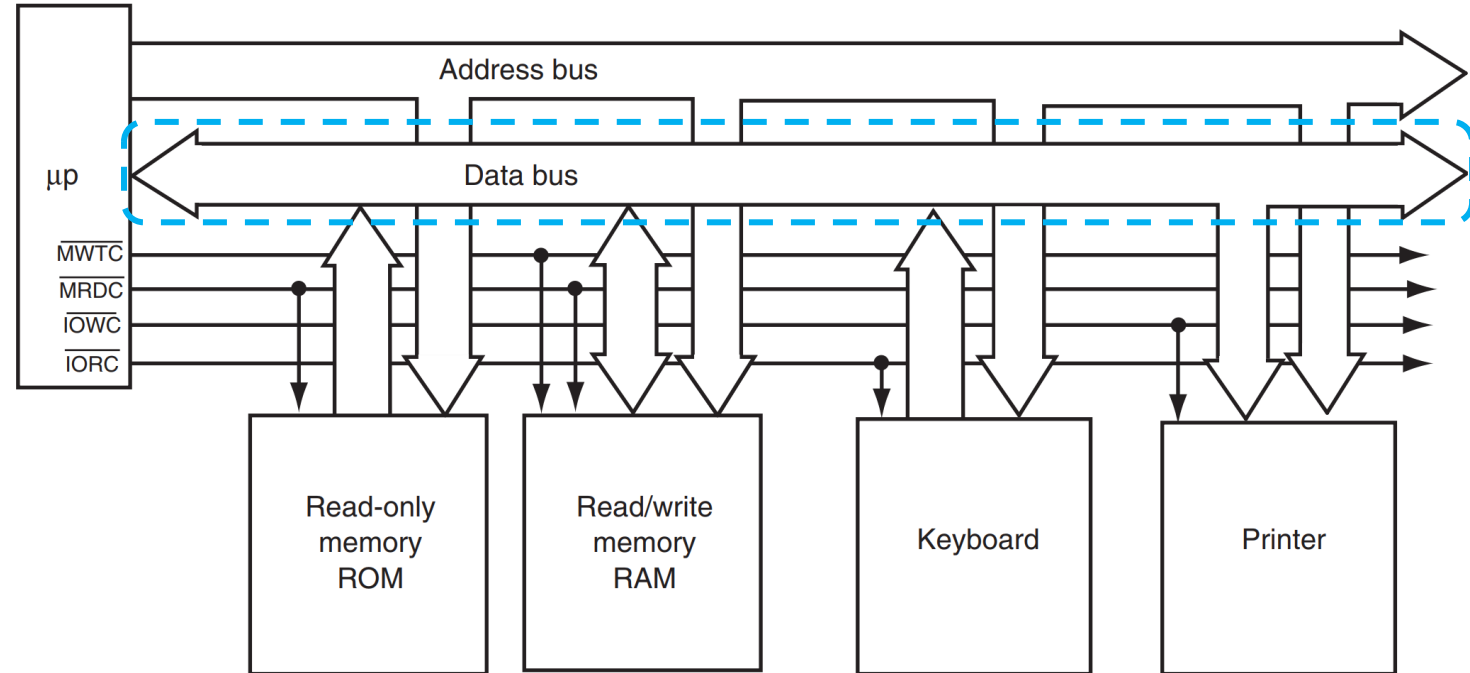
- It is a *unidirectional* bus.
- Information transfer takes place from the μP to the **memory** or **I/O** elements.
- Typically, 16, 20, 24, 32 or 36 bits long.
- On these lines the CPU sends out the address of the memory location or I/O port that is to be *written to* or *read from*.
- The number of locations that the CPU can address is determined by the number of address lines.
 - ✓ **16 pins:** $2^{16} = 65,536$ bytes
 - ✓ **20 pins:** $2^{20} = 1,048,576$ bytes
 - ✓ **32 pins:** $2^{32} = 4,294,967,296$ bytes



Microcomputer

Data Bus

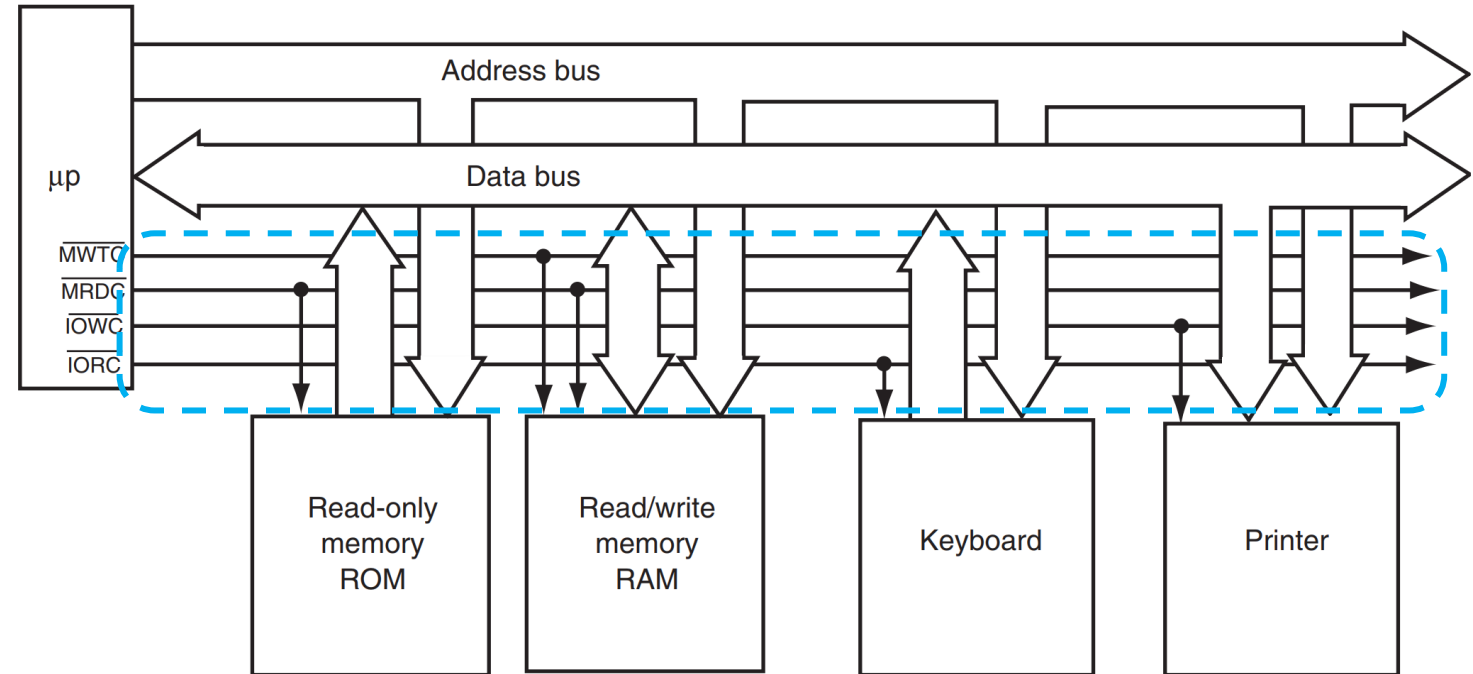
- It is a *bidirectional* bus.
- Data can flow in both directions, that is, to or from the microprocessor.
- The size of the data bus varies from one microprocessor to another.
- Usually matches the word length of the microprocessor.
(*bits* processed in a *single CPU cycle*)
- Usually, a multiple of 8.
- We talk of 4-bit (*nibble*), 8-bit, 16-bit, 32-bit and 64-bit processors which refers to the normal word length of the microprocessor.



Microcomputer

Control Bus

- It consists of a number of signals that are used to synchronize operation of the individual microcomputer elements.
- Consists of potentially many signals.
 - ✓ Read
 - ✓ Write
 - ✓ Read/ $\overline{\text{Write}}$ — Single pin
 - ✓ Interrupt control (**INTR**/**INTA**)
 - ✓ Bus control signals for Direct Memory Access (**HOLD**/**HOLDA**)



Instruction Cycle

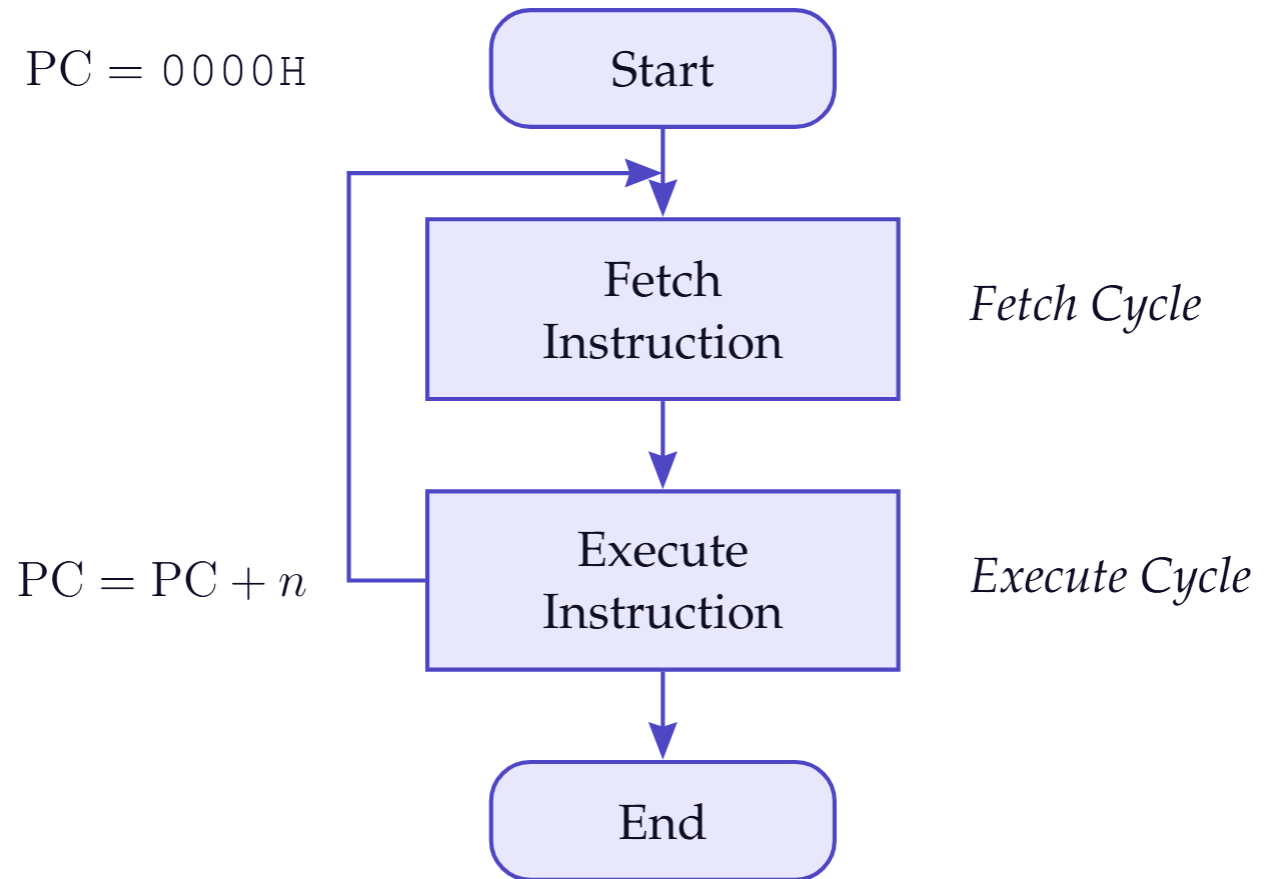
Fetch–Execute Loop

▪ Fetching Cycle

- ✓ The fetch cycle takes the instruction required from memory, stores it in the instruction register, and
- ✓ Moves the program counter on one so that it points to the next instruction.

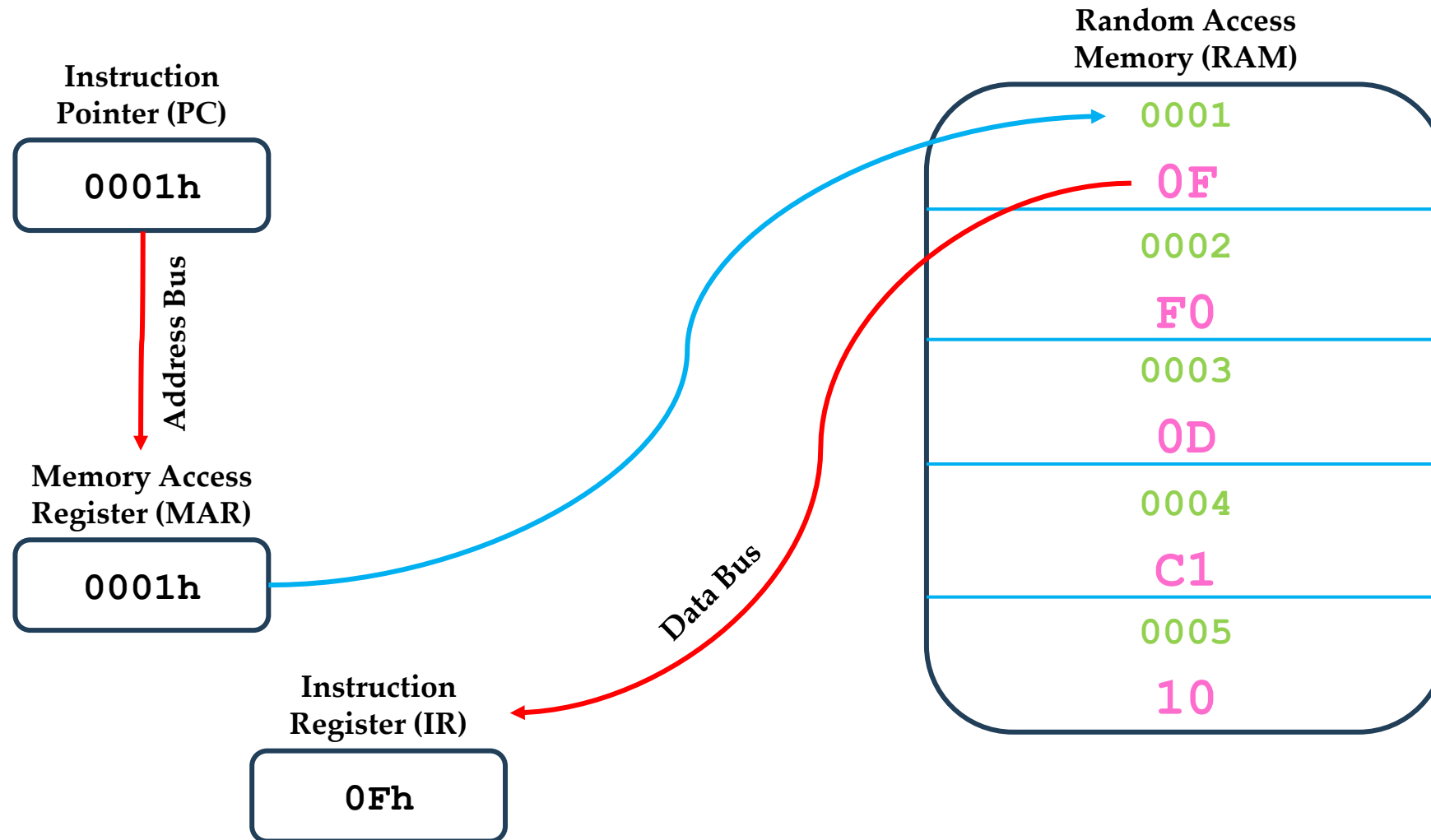
▪ Execution Cycle

- ✓ The actual actions which occur during the execute cycle of an instruction.
- ✓ Depends on both the instruction itself and the addressing mode specified to be used to access the data that may be required.



Fetching Cycle

A simple example simulation



Execution Cycle

A simple 3-instruction program

How this program behaves:

- IP** first points to **0100H** and fetches the byte **E4**. Since **E4** is the opcode for **IN AL, imm8**, the processor knows that the next byte at **0101H** must also be fetched as the operand, which is **05H**. Then the next instruction begins at **0102H**, where the processor fetches **04**; the following byte at **0103H** is its immediate operand **07H**. Finally, the instruction beginning at **0104H** uses **E6** as opcode and **02H** at **0105H** as operand.

Immediate
value

; Memory Address	Content (Hex)	Instruction View	; Meaning
0100:	E4	IN AL, 05H	; Opcode for input from immediate port
0101:	05		; Port address 05H
0102:	04	ADD AL, 07H	; Opcode for add immediate to AL
0103:	07		; Immediate data 07H
0104:	E6	OUT 02H, AL	; Opcode for output to immediate port
0105:	02		; Port address 02H